



ENCAP INVESTMENTS L.P.

IT Business Continuity Report

IT Failover Services from Houston to Microsoft Azure

October 2023

Summary

The purpose of this document is to record the results of EnCap Investment's failover testing. This testing helps ensure that EnCap's IT services continue to be available in the event of a Houston office closure, and that all IT services continue through Microsoft Azure, or vice versa. Microsoft Azure has been configured to serve as a secondary location for IT services along with providing other day-to-day services. In the event of a failover, some of these IT services are moved automatically while others require manual intervention.

Objectives

The principal objective of this exercise is to have users complete a successful test of the failover procedure and verify that, in the event of a physical office closure, they are able to access services in the Azure cloud.

Testing Procedure

IT requested certain users to complete the test procedure:

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IT Services in Scope for Application and Results

VPN

Failover Type: Automatic.

High-Level Process/Info: A VPN is always available to users for either the Houston network or the Azure network.

Test Process: Ensure VPN services are available in Azure by connecting through Azure VPN, and the Houston VPN is available after disconnecting from the Azure VPN.

Results: Success. Both the Azure and the Houston VPNs are independent of each other and always available to users.

Authentication Services Failover

Failover Type: Automatic.

High-Level Process/Info: Connect to EnCap's Azure domain.

Test Process: Users disconnect their mobile devices from the local network, then connect to a non-EnCap Wi-fi network and then authenticate to the Azure network.

Results: Success. Users were able to log into EnCap's Azure domain using the Azure VPN from a non-EnCap Wi-Fi connection.

File Services

Failover Type: Automatic.

High-Level Process/Info: File services and data through DFS-R replication are always synchronized between the local file server (FSHOU03) and the Azure file server (FSAZ01). If the local server is not available then the Azure server becomes the primary file server.

Test Process: Users must observe that, after failover, all file services on the Azure file server are available.

Results: Success. After failover, users were able to verify that file services were available.

IT Services Out of Scope

These services are considered out of scope and not critical for using the Azure services remotely.

Houston Phone System

Reason: Phone systems run independently in Houston.

Backups

Reason: Because backups are uploaded to the Barracuda cloud service, the data is hosted at both the Houston office and in the Barracuda cloud. Azure servers are backed up by Microsoft at geo-redundant locations and run in high availability.

Misc EnCap Applications

Reason: Considered non-critical engineering applications: Spotfire, Di Transform and Aries.

Conclusion

The failover testing was successful and additional testing will re-commence on the next cycle.